



Sink or Swim

Effect of Injuries on Player Performance

A comprehensive look at game performance pre and post injury in Basketball



Introduction

Motivation

- With many star players getting injured, we set out to quantify the relationship between in-game performance pre-injury vs post-injury

Past Research

- One study examined the relationship between performance only after an Achilles tear (Khalil et al., 2020)
- Another study examined how NFL players performed post ACL injury (Mody et al., 2022)





The “Big Question”

What impact do injuries have on future player performance,
and how should GMs treat players with injuries?

Null Hypothesis

Injuries do not significantly affect
in-game performance

Research Hypothesis

Injuries significantly decrease
player performance; therefore their
BPM will decrease



Background Info: Stats and Injuries

Statistics

BPM (Box Plus Minus)

Rank	Player	BPM	Season
1.	Nikola Jokić	13.72	2021-22
2.	Nikola Jokić	13.28	2024-25
3.	LeBron James	13.24	2008-09
4.	Nikola Jokić	13.23	2023-24
5.	Nikola Jokić	13.01	2022-23
6.	Michael Jordan *	12.96	1987-88
7.	Nikola Jokić	12.09	2020-21
8.	Michael Jordan *	12.01	1990-91
9.	Stephen Curry	11.94	2015-16
10.	Michael Jordan *	11.88	1988-89
11.	David Robinson *	11.87	1993-94
12.	LeBron James	11.81	2009-10
13.	LeBron James	11.71	2012-13

Common Basketball Injuries

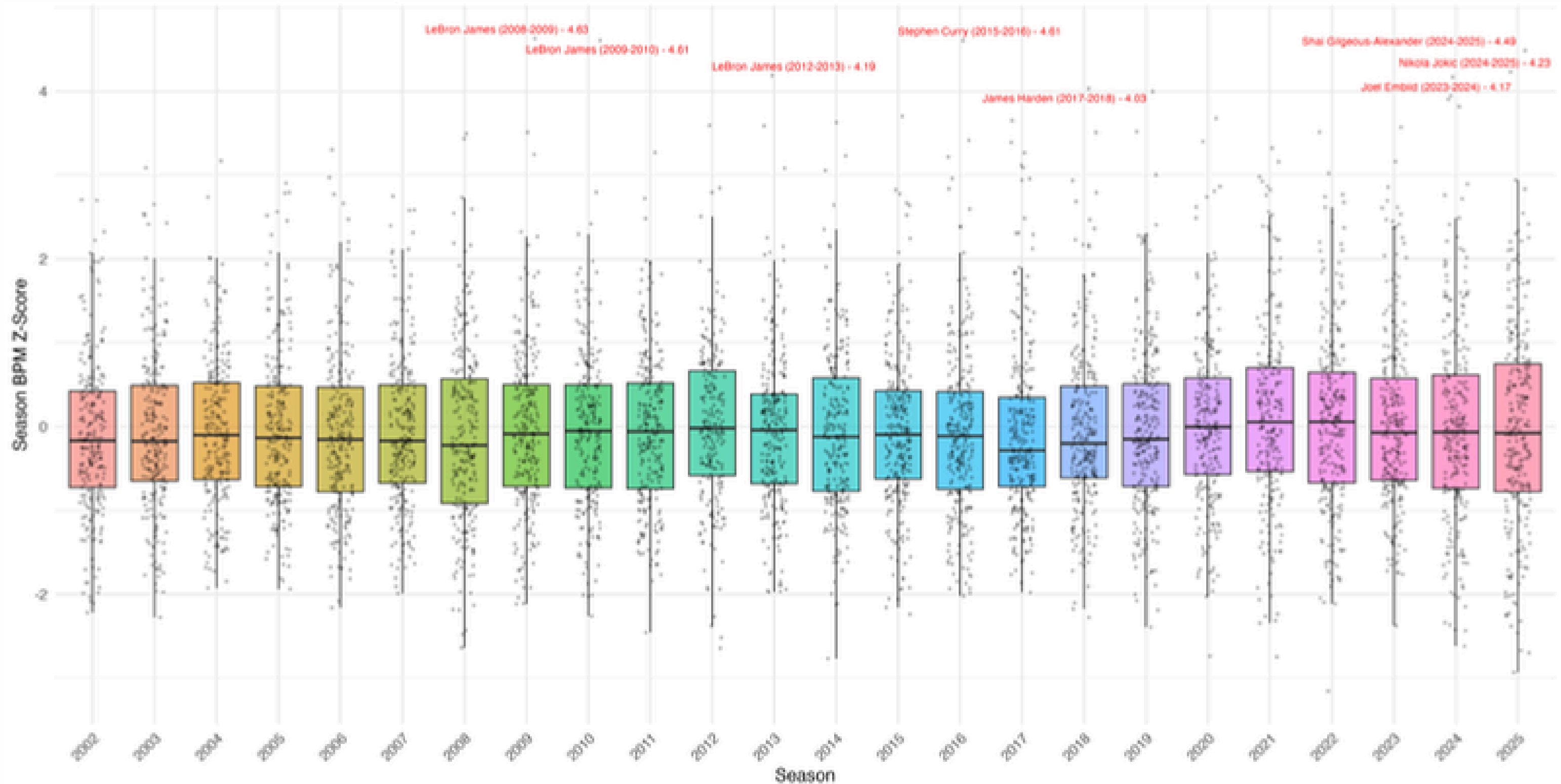
Soft tissue pulls, strains, tears



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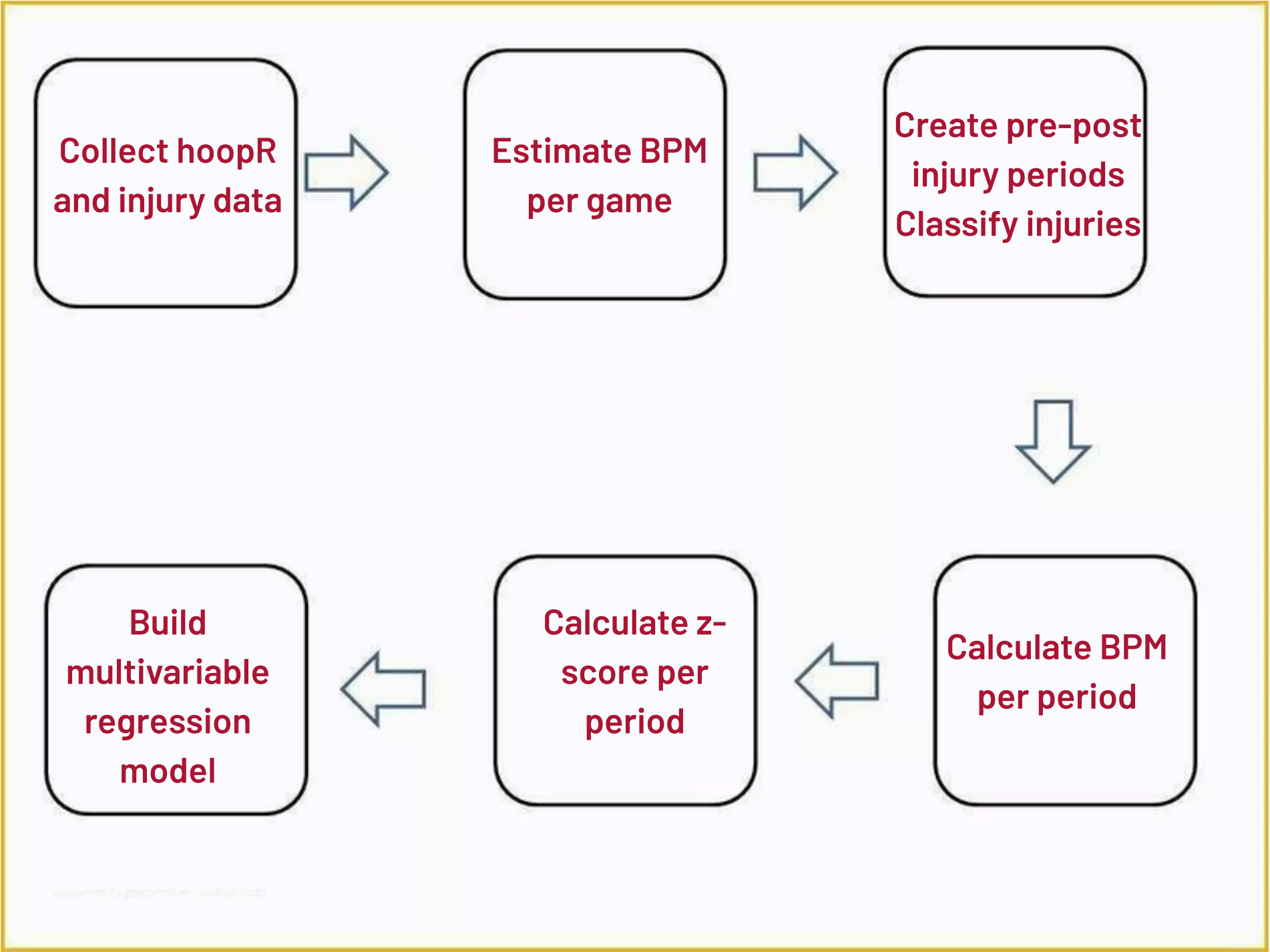


Season BPM Z-Score Distribution





Our Procedure



```
# --- Creating periods ---

# 1. Comeback events
comebacks <- injuries %>%
  filter(!is.na(Acquired)) %>%
  select(player = Acquired, event_date = Date) %>%
  mutate(event_type = "comeback")

# 2. Season starts
season_starts <- df %>%
  arrange(player, season, game_date) %>%
  group_by(player, season) %>%
  slice(1) %>%
  ungroup() %>%
  select(player, season, event_date = game_date) %>%
  mutate(event_type = "season_start")

# 3. Combine events and assign period numbers
events <- bind_rows(season_starts %>% select(player, event_date, event_type),
  comebacks %>% select(player, event_date, event_type))

mutate(event_date = as.Date(event_date)) %>%
  arrange(player, event_date) %>%
  group_by(player) %>%
  mutate(period = row_number()) %>%
  ungroup()

# 4. Assign period to each game
df <- df %>%
  mutate(game_date = as.Date(game_date)) %>%
  arrange(player, game_date) %>%
  left_join(events %>% select(player, event_date, period),
    by = c("player" = "player", "game_date" = "event_date")) %>%
  group_by(player) %>%
  fill(period, .direction = "down") %>%
  ungroup()
```

```
)

player_season_stats <- player_season_stats %>%
  mutate(
    pts_per_tsa = points / pmax(field_goals_attempted + 0.44 * free
  )

team_pts_per_tsa <- player_season_stats %>%
  group_by(team_season_id) %>%
  summarise(team_pts_per_tsa = sum(points) / sum(field_goals_attempted))

player_season_stats <- left_join(player_season_stats, team_pts_per_tsa)

player_season_stats <- player_season_stats %>%
  mutate(
    threshold = team_pts_per_tsa - 0.33,
    threshold_pts = (pts_per_tsa - threshold) * (field_goals_attempted + 0.44 * free_throw_attempts)
  ) %>%
  mutate(threshold_pts = pmax(threshold_pts, 0))

team_threshold_pts <- player_season_stats %>%
  group_by(team_season_id) %>%
  summarise(team_threshold_pts = sum(threshold_pts), .groups = "drop")

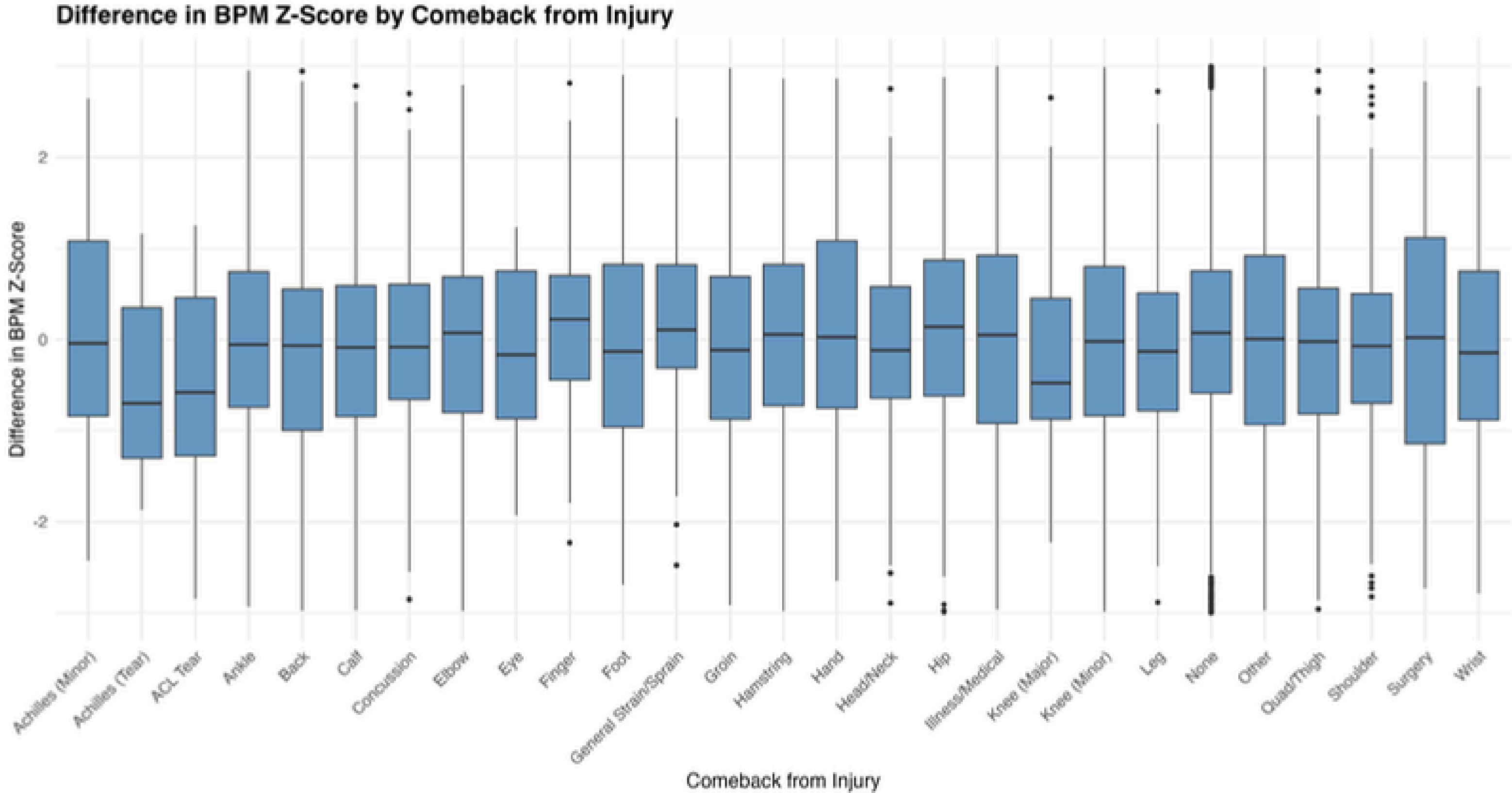
player_season_stats <- left_join(player_season_stats, team_threshold_pts)

player_season_stats <- player_season_stats %>%
  mutate(threshold_pts_pct = threshold_pts / ifelse(team_threshold_pts > 0, team_threshold_pts, 1))

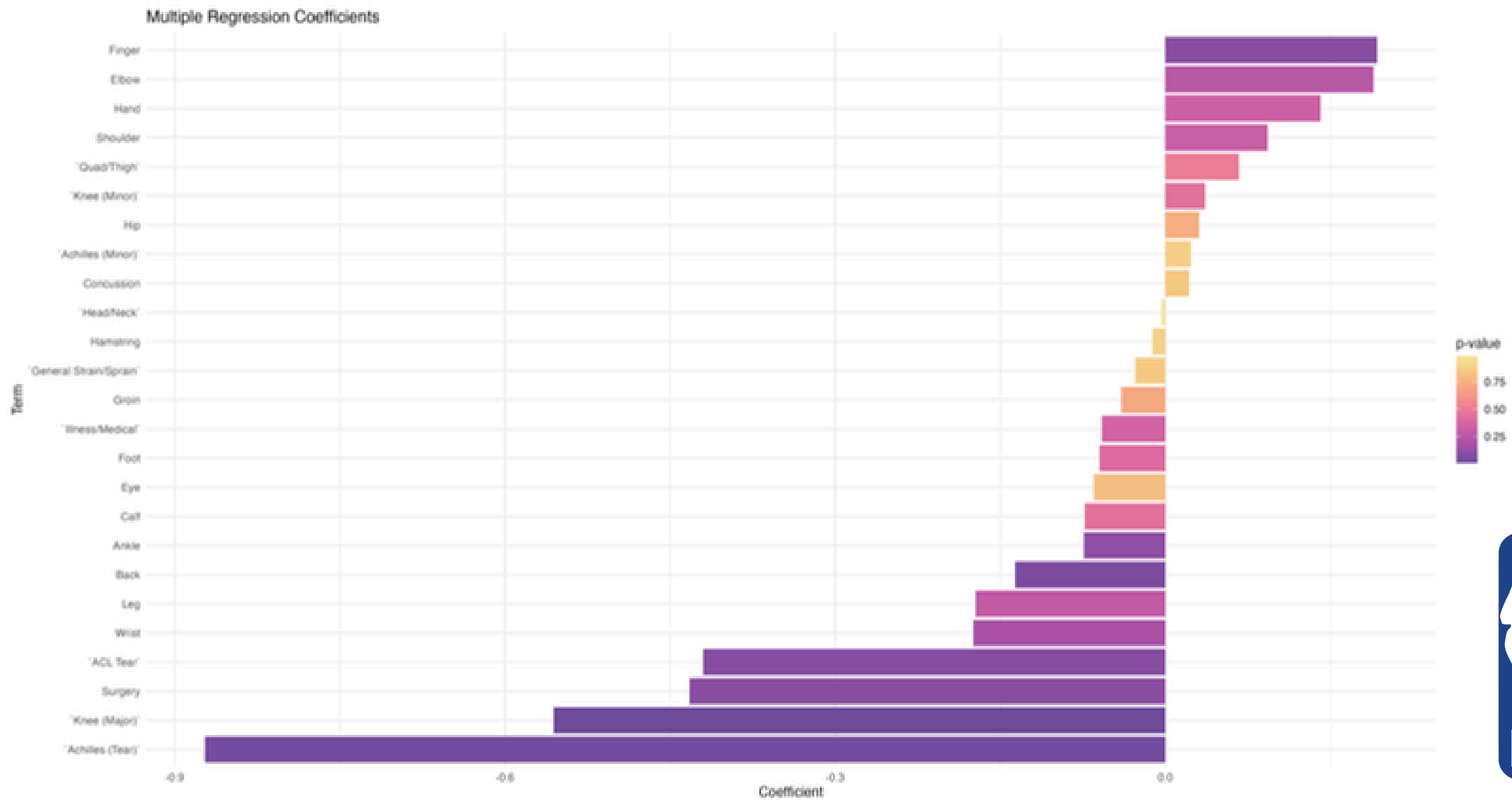
# Estimate position (regression)
pc <- self$position_coeffs
player_season_stats <- player_season_stats %>%
  mutate(
    estimated_position =
      pc$intercept +
      pc$trb_pct * trb_pct +
      pc$stl_pct * stl_pct +
      pc$pf_pct * pf_pct +
      pc$ast_pct * ast_pct +
      pc$blk_pct * blk_pct
  )
```



Difference in BPM Score by Injury



Regression Coefficients by Injury



Haliburton, Tatum, Irving

Who will have a greater bounceback?



Player	Tyrese Haliburton	Jayson Tatum	Kyrie Irving
Age	26	28	33
Position Group	Small	Big	Small
Injury	Achilles (Tear)	Achilles (Tear)	ACL Tear
Previous Season Z-Score	2.4176	2.5480	1.7209
Predicted Z-Score	-0.0870	-0.0639	0.2663
Predicted Difference	-2.5046	-2.6119	-1.4546



Conclusion

Our Findings

Major injuries, primarily to the ACL and Achillies significantly decrease player performance

Minor injuries do not significantly impact player performance

Why is it important?

Helps fans and coaches understand load management

Helps GMs understand how to build around stars who got injured



Our Faults and Limitations

Survivorship Bias

Regression Fallacy

Recovery Time Variance

Biomechanical Advancements



References

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Questions?